Install and manage packages

The recommended way to install Debian packages (“deb” files) is using the Advanced Packaging Tool (APT), which can be used on the command line using the `apt` utility.

The commands contained within `apt` provide the means to install new software packages, upgrade existing software packages, update the package list index, and even upgrade the entire Ubuntu system.

Update the package index

The APT package index is a database of available packages from the repositories defined in the `/etc/apt/sources.list.d` directory. Ubuntu repositories are defined in the `/etc/apt/sources.list.d/ubuntu.sources` file. To update the local package index with the latest changes made in the repositories, and thereby access the most up-to-date version of the package you’re interested in, run the following:

```
$ sudo apt update
```

Note

Ubuntu releases prior to 24.04 LTS (Noble), which don’t use the deb822 standard by default, have repository information defined in the `/etc/apt/sources.list` file.

Install a package

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```
$ sudo apt remove nmap
```

Adding the `--purge` option to `apt remove` will remove the package configuration files as well. This may or may not be what you want, so use it with caution.

> **Note**
>
> While `apt` is a command-line tool, it is intended to be used interactively, and not to be called from non-interactive scripts. The `apt-get` command should be used in scripts (perhaps with the `--quiet` flag). For basic commands the syntax of the two tools is identical.

Upgrading packages

Installed packages on your computer may periodically have upgrades available from the package repositories (e.g., security updates). To upgrade your system, first update your package index and then perform the upgrade – as follows:

```
$ sudo apt update
$ sudo apt upgrade
```

For details on how to upgrade to a new Ubuntu release, see our guide on upgrading releases. For further information about using APT, read the comprehensive APT User's Guide, or type `apt help`.

Aptitude

Launching Aptitude with no command-line options will give you a menu-driven, text-based frontend to the APT system. Many of the common package management functions, such as installation, removal, and upgrade, can be performed in Aptitude with single-key commands, which are typically lowercase letters.

Aptitude is best suited for use in a non-graphical terminal environment to ensure the command keys

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When Aptitude starts, you will see a menu bar at the top of the screen and two panes below the menu bar. The top pane contains package categories, such as “New Packages” and “Not Installed Packages”. The bottom pane contains information related to the packages and package categories.

Using Aptitude for package management is relatively straightforward thanks to its user interface. The following are examples of common package management functions as performed in Aptitude:

Installing packages

To install a package, locate it in the “Not Installed Packages” category by using the keyboard arrow keys and the `Enter` key.

Highlight the desired package, then press the `+` key. The package entry should turn **green**, which indicates it has been marked for installation. Now press `g` to be presented with a summary of package actions. Press `g` again, and the package will be downloaded and installed. When finished, press `Enter` to return to the menu.

Remove Packages

To remove a package, locate it in the “Installed Packages” category by using the keyboard arrow keys and the `Enter` key.

Highlight the package you want to remove, then press the `-` key. The package entry should turn **pink**, indicating it has been marked for removal. Now press `g` to be presented with a summary of package actions. Press `g` again, and the package will be removed. When finished, press `Enter` to return to the menu.

Updating the package index

To update the package index, press the `u` key.

Upgrade packages

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The first column of information displayed in the package list (in the top pane) lists the current state of the package (when viewing packages). It uses the following key to describe the package state:

- **i** : Installed package
- **c** : Package not installed, but package configuration remains on the system
- **p** : Purged from system
- **v** : Virtual package
- **B** : Broken package
- **u** : Unpacked files, but package not yet configured
- **C** : Half-configured - configuration failed and requires fix
- **H** : Half-installed - removal failed and requires a fix

To exit Aptitude, simply press the `q` key and confirm you want to exit. Many other functions are available from the Aptitude menu by pressing the `F10` key.

Command-line Aptitude

You can also use Aptitude as a command-line tool, similar to `apt`. To install the `nmap` package with all necessary dependencies (as in the `apt` example), you would use the following command:

```
$ sudo aptitude install nmap
```

To remove the same package, you would use the command:

```
$ sudo aptitude remove nmap
```

Consult the *Aptitude manual pages* for full details of Aptitude's command-line options.

dpkg

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```
$ dpkg -l
```

Depending on the number of packages on your system, this can generate a large amount of output. Pipe the output through `grep` to see if a specific package is installed:

```
$ dpkg -l | grep apache2
```

Replace `apache2` with any package name, part of a package name, or a regular expression.

List files

To list the files installed by a package, in this case the `ufw` package, enter:

```
$ dpkg -L ufw
```

If you are unsure which package installed a file, `dpkg -S` may be able to tell you. For example:

```
$ dpkg -S /etc/host.conf
base-files: /etc/host.conf
```

The output shows that the `/etc/host.conf` belongs to the base-files package.

> **Note**
>
> Many files are automatically generated during the package install process, and even though they are on the filesystem, `dpkg -S` may not know which package they belong to.

Installing a deb file

You can install a local `.deb` file by entering:

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> **Caution**
>
> Uninstalling packages using `dpkg`, is **NOT** recommended in most cases. It is better to use a package manager that handles dependencies to ensure that the system is left in a consistent state. For example, using `dpkg -r zip` will remove the `zip` package, but any packages that depend on it will still be installed and may no longer function correctly as a result.

For more `dpkg` options see the *dpkg(1)* manual page: `man dpkg`.

APT configuration

Configuration of the APT system repositories is stored in the `/etc/apt/sources.list.d` directory. The Ubuntu repositories are stored in `/etc/apt/sources.list.d/ubuntu.sources`. You can edit this file to enable and disable repositories.

Automatic updates

It's possible to setup an Ubuntu system with Automatic Updates, such that certain types of upgrades are applied automatically. In fact, the default for Ubuntu Server is to automatically apply security updates. Please see the Automatic updates section for details.

Extra repositories

In addition to the officially-supported package repositories available for Ubuntu, there are also community-maintained repositories which add thousands more packages for potential installation. Two of the most popular are the *Universe* and *Multiverse* repositories. These repositories are not officially supported by Ubuntu, but because they are maintained by the community they generally provide packages which are safe for use with your Ubuntu computer.

For more information, see our guide on using third-party repositories.

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Many other package sources are available – sometimes even offering only one package, as in the case of packages provided by the developer of a single application. You should always be cautious when using non-standard package sources/repos, however. Research the packages and their origins carefully before performing any installation, as some packages could render your system unstable or non-functional in some respects.

By default, the *universe* and *multiverse* repositories are enabled. If you would like to disable them, edit `/etc/apt/sources.list.d/ubuntu.sources` and remove `universe` and `multiverse` from `Components`:

```
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: DISTRO-SHORT-CODENAME DISTRO-SHORT-CODENAME-updates DISTRO-SHORT-CODEN
Components: main universe restricted multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

Logging

Actions of the `apt` command, such as installation and removal of packages, are logged in the `/var/log/dpkg.log` log file.

Further reading

Most of the material covered in this chapter is available in the respective man pages, many of which are available online.

- The Installing Software Ubuntu wiki page has more information.
- The APT User's Guide contains useful information regarding APT usage.

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